

Reading the Confusion Matrix

Accuracy, precision, recall and F1 all come out of one small square with four numbers in it. This chapter builds that square from eight passengers, then shows what each of the four numbers asks, where each one fails, and how to choose between them.

A model that predicts survival on a sinking ship will get some of its predictions right and some of them wrong. That much is obvious. What is far less obvious is how to say, in a single number, how well it did.

Most people reach for the same number first. Count the ones it got right, divide by the total. It is a reasonable instinct and it is the first thing anybody invents. It is also the number that will, on the most important problems you are likely to meet, tell you a model is excellent when it is useless.

This chapter works through the alternatives. It uses eight passengers throughout, few enough to count on your fingers, and every number in it is worked out from those eight. Nothing is quoted from a benchmark. By the end you will be able to read any of these four measures off a single square, and, more importantly, to say which one your own problem actually needs.

A guess and an answer are two different things

You have eight predictions and eight answers, and no way yet to turn them into a number anyone can act on. Start with what you actually hold. Eight people were aboard. Some survived and some did not, and that is settled history: we have the record. Separately, a computer program looked at each person's ticket class, age and fare and guessed whether they made it.

The program is the **model**. Each guess it made is a **prediction**. The historical record of who actually survived is the **actual label**. These two get muddled constantly, so hold them apart from the start. The actual label is what happened. The prediction is what the model said would happen. The model does not get a vote on the first one.

Judging the model means laying its eight guesses next to the eight answers and looking at where they disagree. Every figure in this chapter draws each person as one small square carrying both facts at once, so the encoding is worth learning now. The inside of the square is the truth: a solid filled middle means the person survived, and an empty middle means they did not. The ring drawn around it is the guess: a heavy unbroken ring means the model predicted survival, and a thin broken ring means it predicted death. The middle is a filled area because it is a fact about the world that is simply there. The ring is an outline around that fact because a guess is laid on top of the truth from outside.

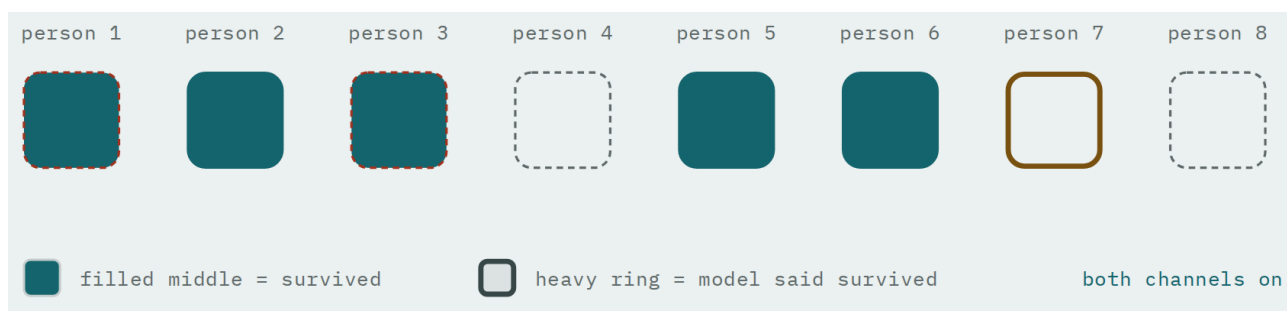


Figure 1.1 The eight passengers, each drawn with the recorded outcome as the fill and the model's prediction as the ring.

Figure 1.1 shows all eight. Persons 2, 5 and 6 have a filled middle and a heavy ring: they survived, and the model said so. Persons 1 and 3 have a filled middle but only a thin broken ring. Those two survived and the model missed them.

Because two facts are carried in one shape, either can be read on its own. Figure 1.2 switches the ring off and leaves only the record, which is the column a historian would have. Figure 1.3 does the reverse, leaving only what the model claimed.



Figure 1.2 The same eight with the ring switched off, showing only what actually happened.

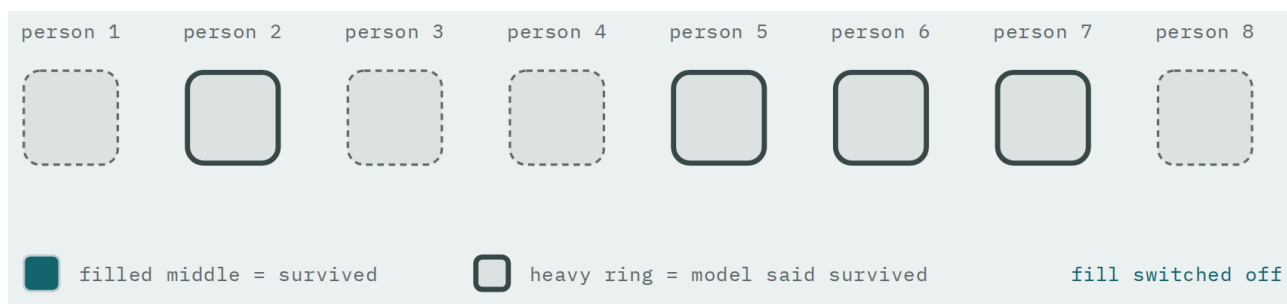


Figure 1.3 The same eight with the fill switched off, showing only what the model predicted.

Five people survived. The model predicted survival for four. Those two counts are not the same, and they are not even counting the same thing, which is the first place this subject trips people. One is a fact about the ship. The other is a claim about the model.

A teacher marking multiple-choice papers holds the answer key in one hand and a pupil's sheet in the other, and for each question does exactly one thing: compares two marks. She never asks whether the pupil's answer was reasonable or nearly right. The only material fact is whether the two marks match, one question at a time. Every measure in this chapter is a different way of totting up those matches and mismatches at the end.

Before those mismatches can be counted, though, one word has to be pinned down. The four names everyone uses depend on it entirely, and it does not mean what it sounds like.

Positive means the thing you went looking for

You now have a way to see agreement and disagreement, and no vocabulary for either. The vocabulary depends on one word, and that word is a trap. Answer quickly, before reading on: in a test for a disease, which result is the positive one?

Most people say the healthy one, because positive sounds like good news. It is the opposite. A positive test result means the disease was found. Positive here has nothing to do with desirable. It marks **the thing being hunted for**, and the thing being hunted for is usually the bad thing: the tumour, the fraud, the fault, the spam.

So the **positive class** is whichever outcome you decided to detect, and the **negative class** is everything else. That decision is yours. It is not a property of the data, and nothing stops you making the other choice. Figure 1.4 counts survival as the positive class and finds five positives; Figure 1.5 counts death instead and finds three.

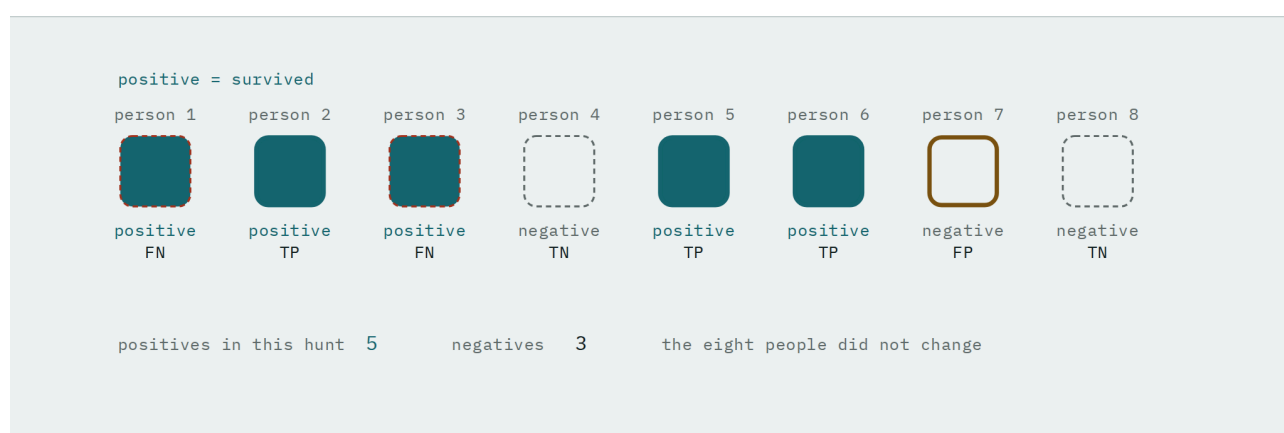


Figure 1.4 The eight passengers with survival counted as the positive class. Five positives.

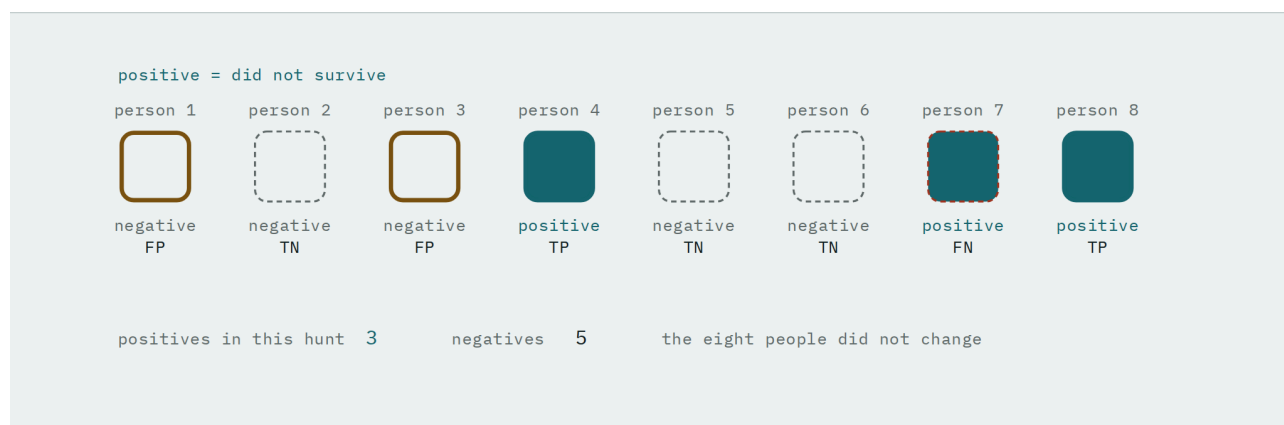


Figure 1.5 The same eight people with death counted as the positive class. Three positives.

Notice what stays put. Person 4 is the same person, with the same history and the same guess against their name, in both figures. The model has not been retrained and no data has changed. All that moved is which of the two outcomes was named as the one being counted, and the number of positives went from five to three.

This matters more than it looks, because every number in the rest of this chapter is computed with respect to the positive class. If you report a recall of 0.60 without saying what counted as positive, you have reported nothing at all. In a medical paper that is the difference between finding sixty

percent of the sick people and finding sixty percent of the healthy ones.

A metal detector on a beach behaves the same way. Sweeping for coins, every beep is a result worth having and every bottle cap is a nuisance. Sweeping to clear the sand of bottle caps, the beeps mean the opposite. The detector is unchanged and the beach is unchanged. Only the thing you named as the target has moved, and with it every count you keep in your notebook.

For the rest of this chapter, positive means survived. Each of the eight comparisons can now be given its proper name.

The second word is the guess, the first says whether it held

With the target named, each of the eight comparisons can be sorted. Exactly four things can happen when a yes-or-no guess meets a yes-or-no answer. Two are the model being right. Two are the model being wrong, and those two are not the same kind of wrong at all.

The names look forbidding and they are built from a rule that takes about ten seconds to learn. Read each name backwards.

The second word is what the model said. A **true positive** and a **false positive** are both cases where the model said positive. The first word says whether that call held up: true means it was correct, false means it was not. So a **false negative** is not a double negative to untangle. It simply means the model said negative and was wrong to. The person survived and the model missed them. A **true negative** is the quiet case where the model said negative and was right.

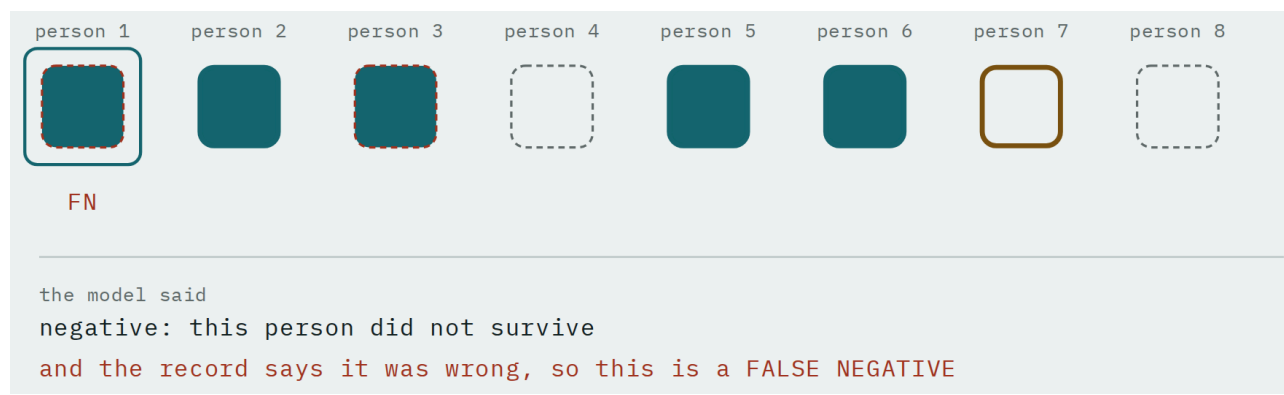


Figure 1.6 Person 1 read out in full: the model predicted death, the record says otherwise, so the case is a false negative.

Figure 1.6 works through a single case. Person 1 survived, the model said they did not, and the name follows mechanically from those two facts. The same procedure applied to all eight gives Figure 1.7.



Figure 1.7 All eight passengers labeled. Three true positives, two false negatives, one false positive, two true negatives.

Three true positives, two false negatives, one false positive, two true negatives. Three plus two plus one plus two is eight, and it always will be. Every case lands in exactly one of the four boxes, because the two questions behind them have two answers each.

A smoke alarm in a kitchen shows why the two kinds of error need separate names. It goes off and there is a fire: the alarm doing its job. It goes off because you burnt the toast: annoying and harmless. It stays quiet on a calm evening: the ordinary case nobody notices. And it stays quiet while the pan is alight. Both mistakes are the alarm being wrong, and only one of them burns the house down.

Hold on to that, because it is the reason this chapter has four measures rather than one. A false positive costs an unnecessary check. A false negative costs the thing you were trying to catch. Any summary that adds them together has already thrown away the distinction.

Eight names in a row is still a list. The next section turns it into a shape you can read at a glance.

Two questions, two answers each, and the square draws itself

Eight names in a row is hard to hold in your head. There is a much better arrangement, and it is not something anyone invented so much as something that falls out of sorting the same eight chips twice. Figures 1.8 to 1.12 follow the sort.

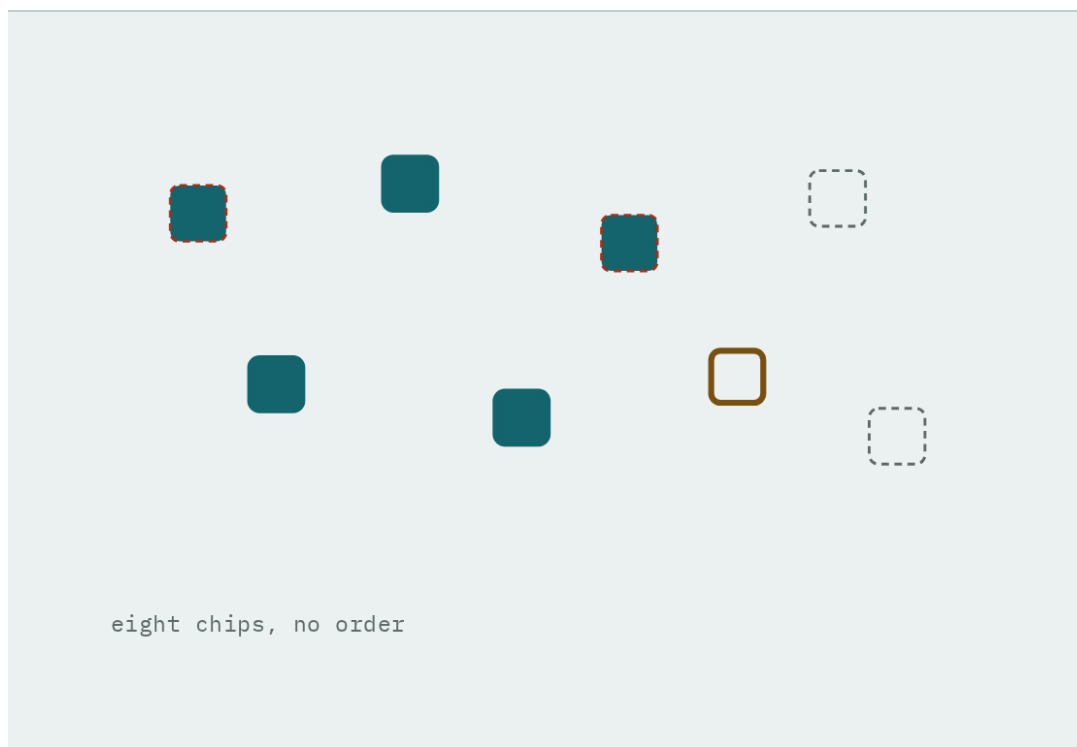


Figure 1.8 The eight chips before any sorting. Each carries its truth and its guess, and the pile says nothing at a glance.

Nothing is added from here on. The same eight chips are only ever moved.

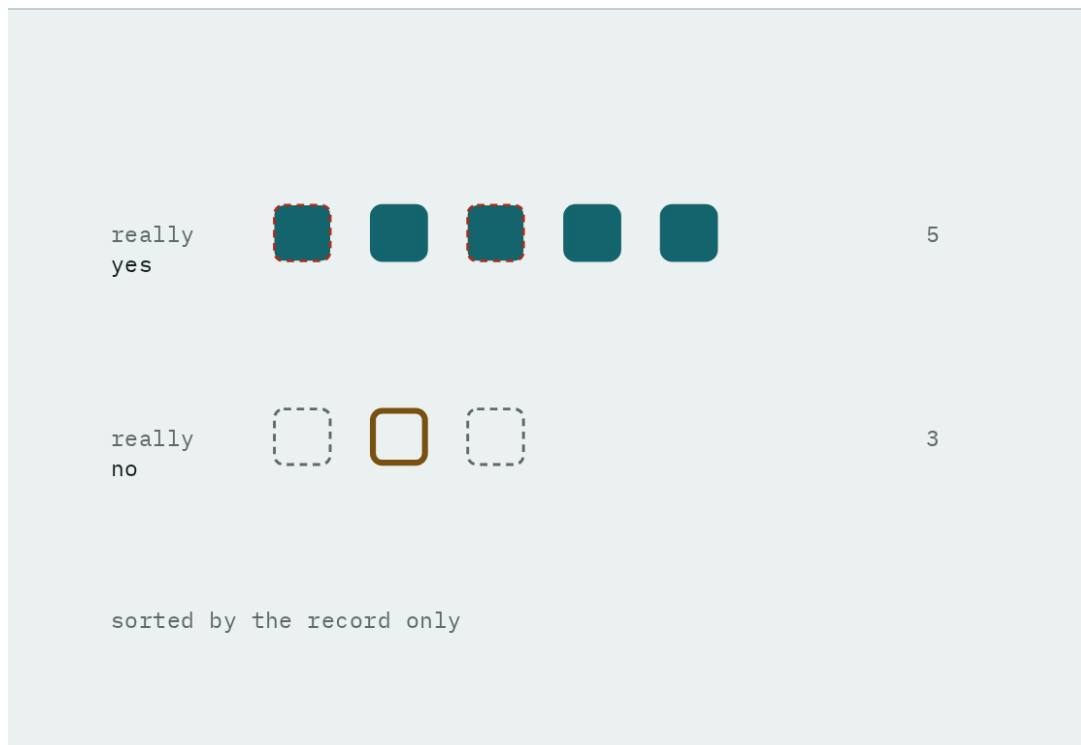


Figure 1.9 The chips split into two rows by the recorded outcome: five survivors above, three deaths below.

The first cut uses only the record. The model has no say in it, which is why these two rows are fixed at five and three no matter how good or bad the model turns out to be. The second cut is the model's.

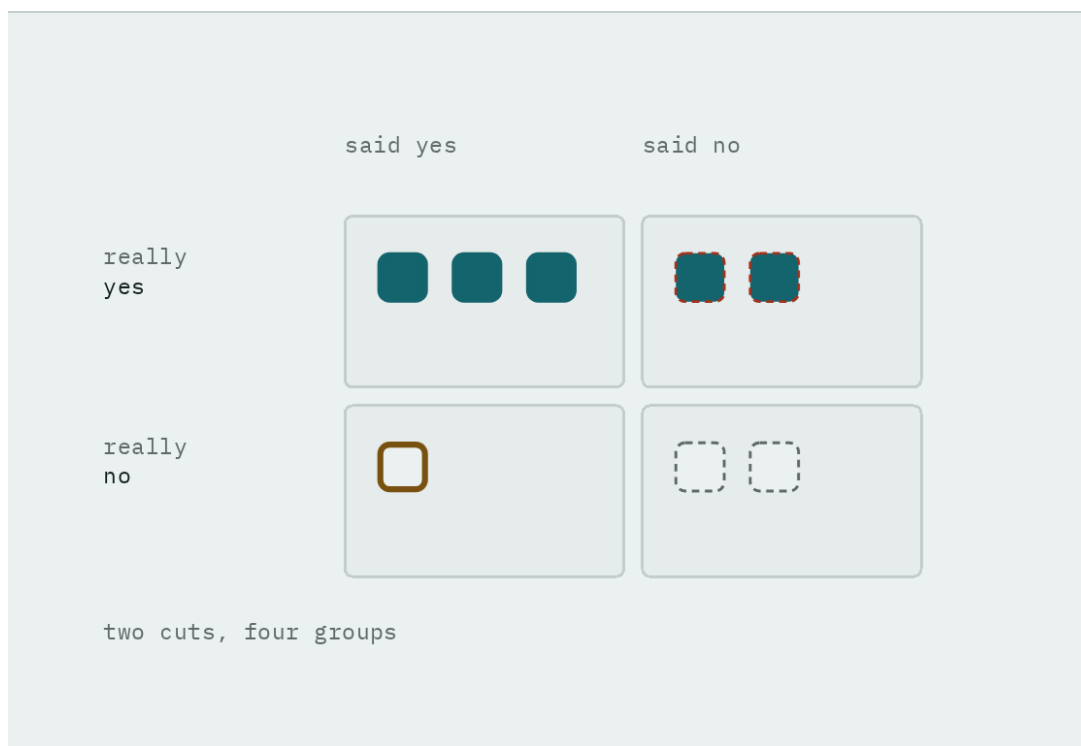


Figure 1.10 Each row split again into columns by what the model predicted. Two cuts, four groups.

Four groups have appeared, and each already has a name from the previous section. Figure 1.11 fills them in.

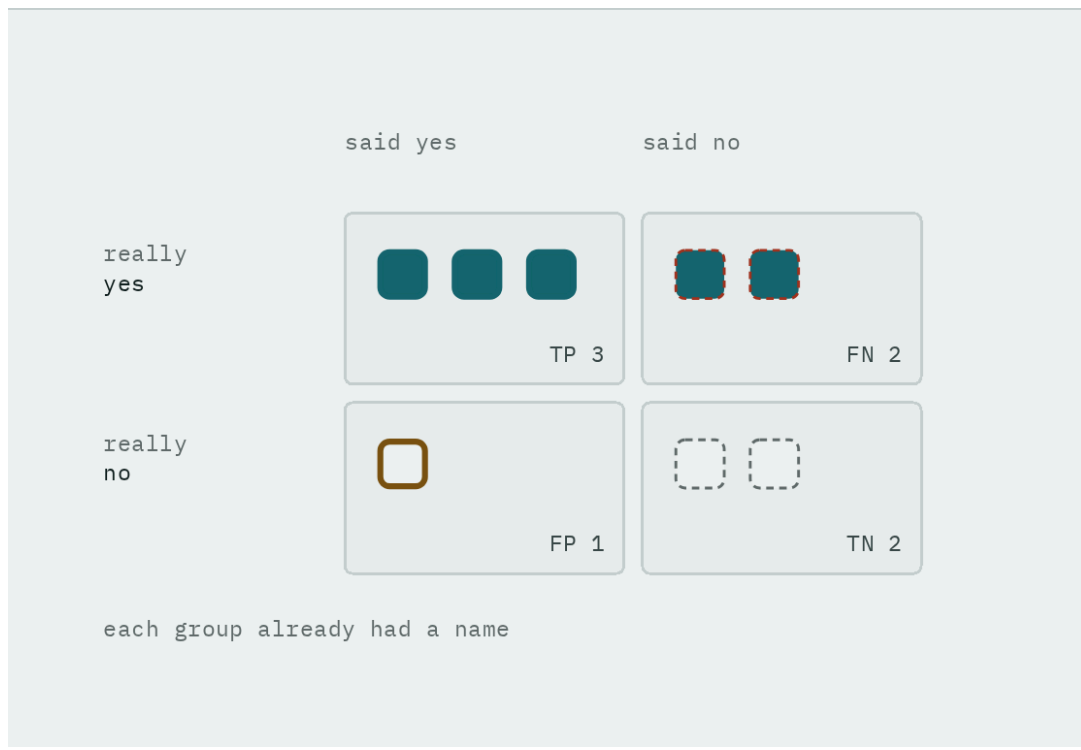


Figure 1.11 The four groups labeled: three true positives, two false negatives, one false positive, two true negatives.

Top left is truth yes and guess yes. Top right is truth yes, guess no. Bottom left is one false positive and bottom right two true negatives. The two boxes running from top left to bottom right are the cases where the model and the record agreed, and it is worth seeing them marked as a set, because the next measure is built from exactly those two.

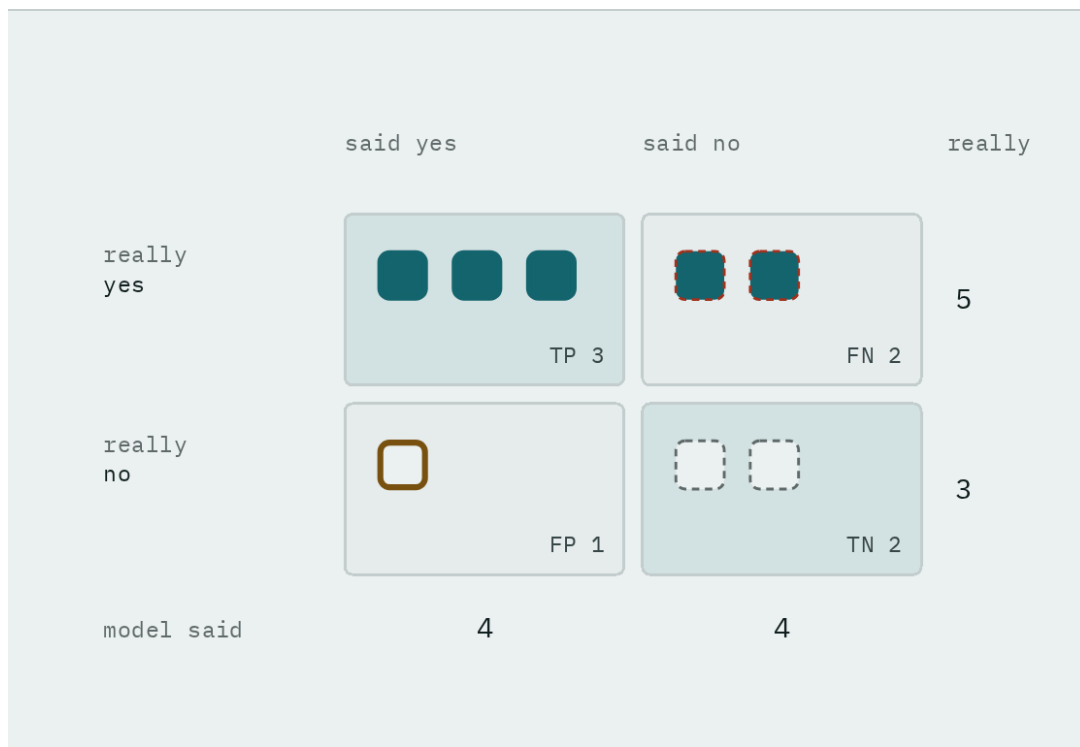


Figure 1.12 Totals added at the edges. The rows sum to five and three, which is history. The columns sum to four and four, which is the model.

That square is the **confusion matrix**. The name is old and means the matrix of confusions: which classes the model mixes up with which. For two classes it is always this shape, always four numbers, and those four numbers contain everything any measure in this chapter will ever use. The original eight people are no longer needed.

Two conventions are worth knowing before you read someone else's work. Some textbooks put predictions down the side and the truth across the top, which flips the whole square. And `scikit-learn`, the library most people meet first, orders its axes negative then positive, so the true negatives land in the top left rather than the bottom right. Always read the axis labels before you read the numbers.

Four numbers, then. The obvious thing to do with them is turn them into a single score, and almost everyone's first attempt is the same one.

Accuracy is the diagonal over the whole square

How often was the model right? Count the cases it got right and divide by the number of cases. That is **accuracy**, and most people invent it without being told it has a name.

On the square, being right is the diagonal: the true positives plus the true negatives. Three plus two is five, out of eight, which is the arrangement in Figure 1.13.

```
accuracy = (TP + TN) / (TP + FP + FN + TN)
          = (3 + 2) / 8
          = 0.625
```

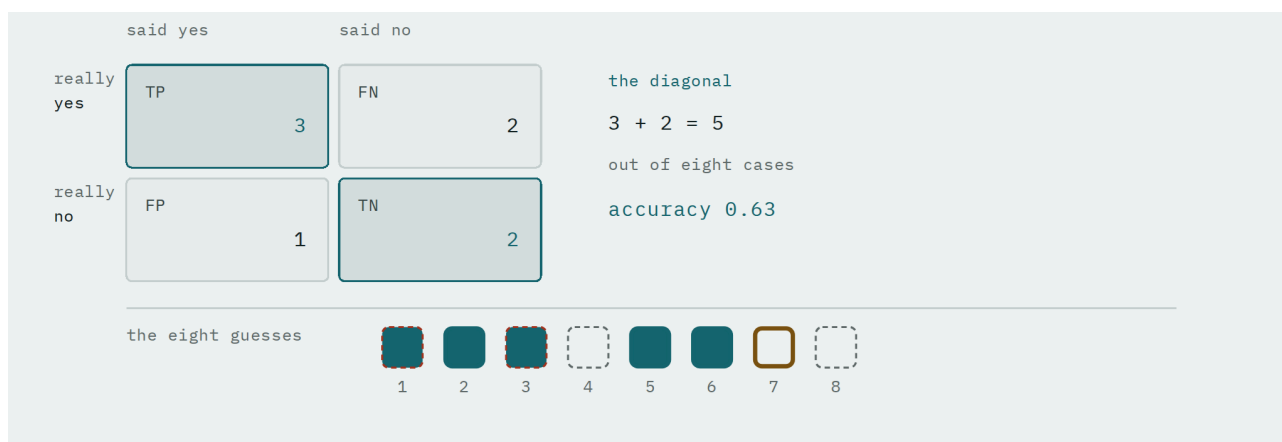



Figure 1.13 The confusion matrix for the model's real predictions, with the diagonal marked and accuracy at 0.625.

Figure 1.14 shows the same square after every prediction is corrected, which is the only arrangement that reaches 1.000. It is included because it is worth seeing what the ceiling looks like: the two off-diagonal boxes empty, and every chip on the diagonal.

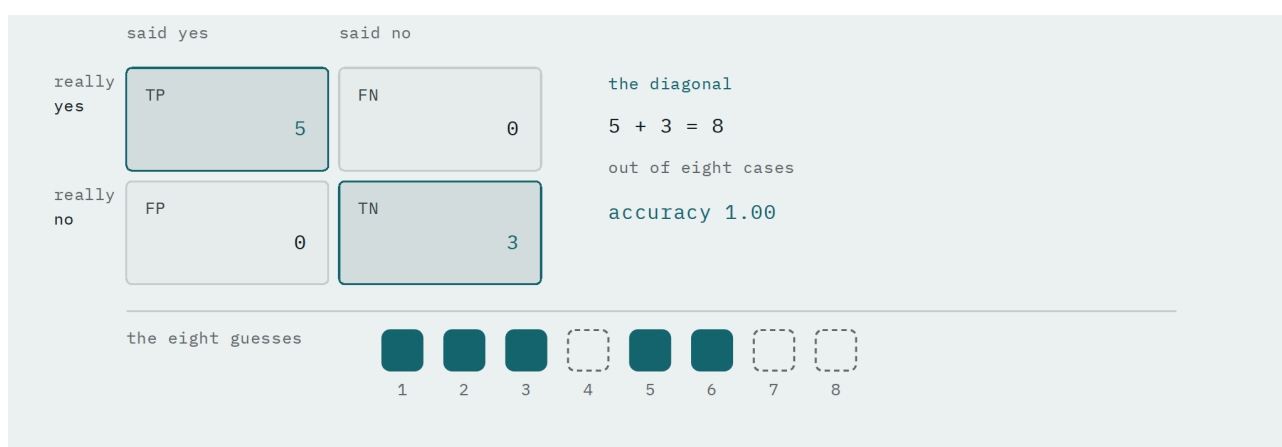


Figure 1.14 The same eight cases with every prediction corrected. The off-diagonal boxes are empty and accuracy reaches 1.000.

Nothing surprising happens between those two figures, and that is exactly why accuracy is so appealing. It behaves the way a score ought to behave, it needs no explanation, and it fits in a headline.

Here the eight cases are reasonably balanced: five survivors and three deaths. Accuracy behaves sensibly on data like that, and if your problem really is that even, it is an honest number to report. Real problems are rarely that even, and once they are lopsided this number stops describing the model at all.

A test that is 99 percent accurate can find nobody

Accuracy behaved well on eight balanced cases, so the question is what happens when the classes are not balanced. This section answers it in five steps, and the answer is worse than most people expect. Leave the ship behind and take a town of a thousand people, ten of whom have an illness worth catching early. Here is a test that costs nothing to build and scores better than almost anything else in this chapter. Figures 1.15 to 1.19 follow it through.



Figure 1.15 One thousand people. Each mark is one person, and the ten who are ill sit apart at the lower right.

The test now runs. It does not look at the patient, it has no inputs, and written down it is one line long.

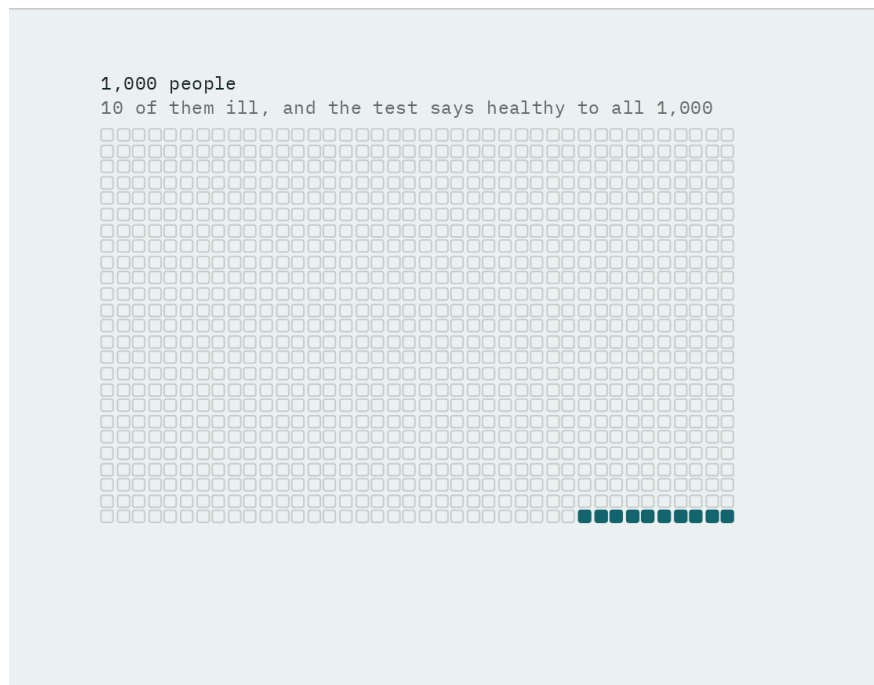


Figure 1.16 The test returns healthy for every one of the thousand people.

It never returns a positive. So every healthy person receives the correct answer, simply because the test's only answer is the one most people deserve.



Figure 1.17 The nine hundred and ninety healthy people, all counted as correct.

Nine hundred and ninety right out of a thousand.



Figure 1.18 Accuracy for the do-nothing test, at 0.990.

That is a higher score than most carefully built models achieve on a hard problem, produced by a test that does nothing whatsoever. And the ten people the test existed to find are still sitting there.



Figure 1.19 None of the ten ill people found. Ten false negatives and not one true positive.

Ten false negatives, no true positives. The one job the test was built for was finding those ten, and it found none of them.

What went wrong is specific. Accuracy counts every case with equal weight, so the nine hundred and ninety easy cases decide the score and the ten that mattered contribute one percent of it. This is **class imbalance**: one class vastly outnumbering the other. It is the normal condition rather than a corner case. Fraud is a fraction of a percent of card transactions. Manufacturing defects are rarer still. Anything worth screening for is, by design, uncommon in the screened population.

The failure gets worse in the direction you would least expect, which Figure 1.20 shows. Make the illness rarer and the useless test scores higher.



Figure 1.20 The same do-nothing test against a rate of one ill person in a thousand. Accuracy rises to 0.999.

At one in a thousand the do-nothing test scores 0.999. The rarer and more serious the thing you are hunting, the more flattering accuracy becomes about a model that never finds it.

A lifeguard who never blows his whistle, judged on how many bathers left the beach safely, has a flawless record on a calm day. The score is dominated by the swimmers who were never in trouble. The only afternoon it would tell you anything about him is the one afternoon it says nothing.

Accuracy asks how often the model was right. On a rare target, that is a question about the easy cases.

The repair follows from the diagnosis. Accuracy pools all four boxes into one fraction, so the enormous true negative box swallows everything. Stop pooling, and ask two narrower questions instead.

One grades your alarms, the other grades your coverage

Two questions, and the whole difference between them is which pile you divide by.

Of everything the model flagged, how much of it was real? That is **precision**. Take the cases the model called positive and ask what fraction genuinely were. It is a question about the model's alarms, and the do-nothing test cannot even be asked it, because it never raised one.

Of everything that was really there, how much did the model find? That is **recall**. Take the cases that genuinely were positive and ask what fraction the model caught. It is a question about coverage, and the do-nothing test scores exactly zero on it. Ten real cases, none found.

On the square, precision reads down a column (Figure 1.21) and recall reads across a row (Figure 1.22). That is the entire distinction, and it is easier to see than to read.

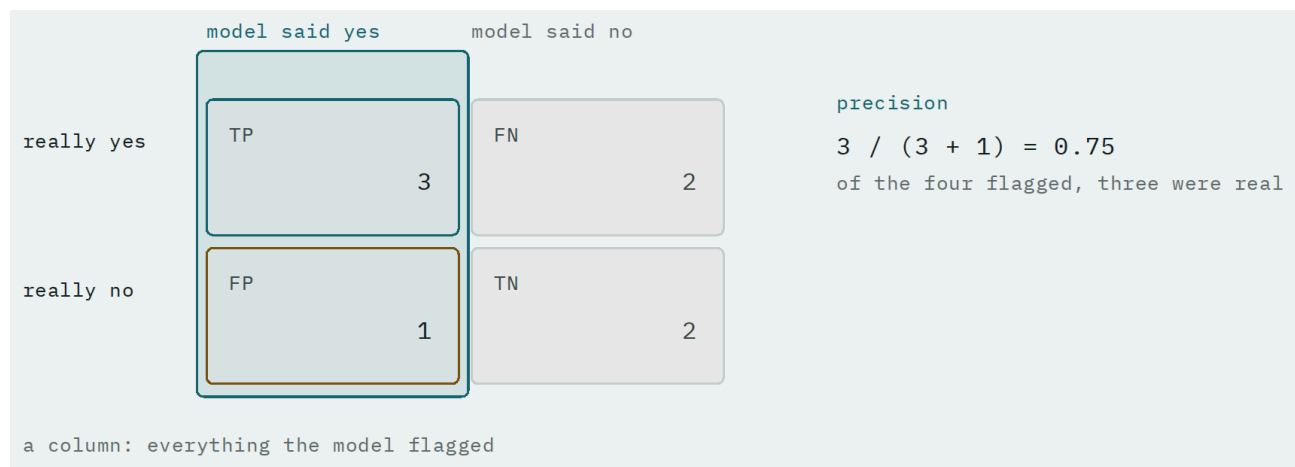


Figure 1.21 Precision reads down the left column: everything the model flagged, three of which were real.

Four people were flagged as survivors and three of them survived, so precision is three quarters.

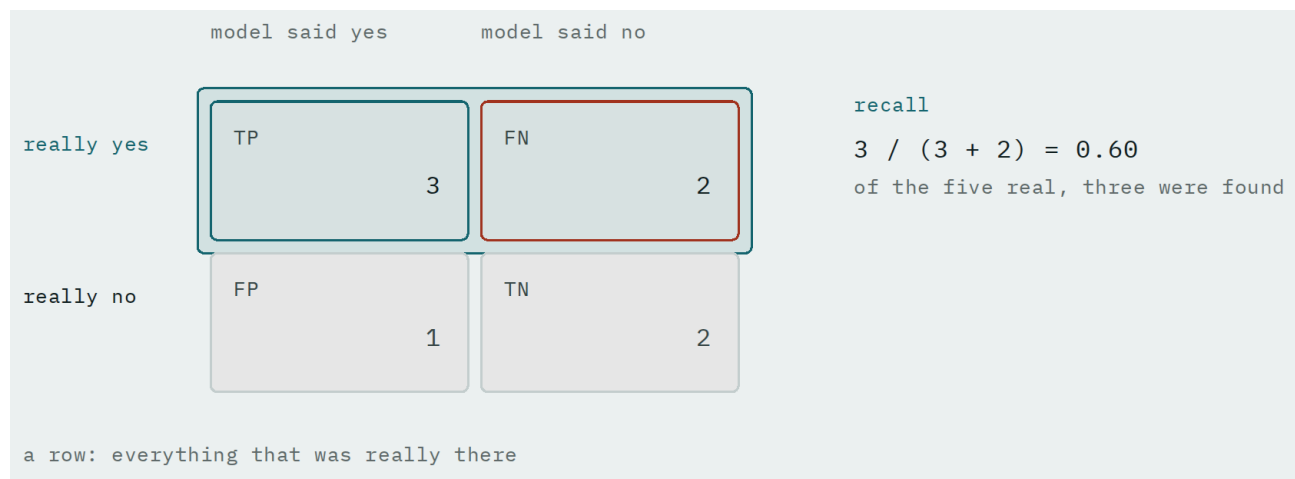


Figure 1.22 Recall reads across the top row: everyone who really survived, three of whom were found.

Five people actually survived and the model found three, so recall is three fifths. Same numerator, different question, different answer.

```
precision = TP / (TP + FP) = 3 / (3 + 1) = 3/4 = 0.75
recall    = TP / (TP + FN) = 3 / (3 + 2) = 3/5 = 0.60
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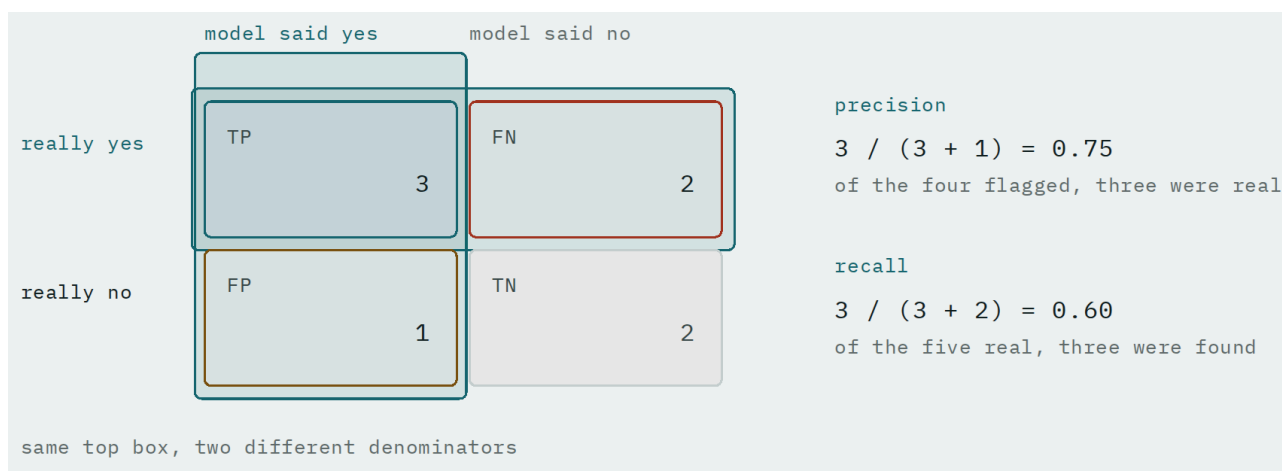


Figure 1.23 Both measures marked together. The true positive box is the numerator of each, and only the denominator moves.

Figure 1.23 makes the shared numerator explicit. Moving the denominator is the whole idea.

A fishing net hauled onto a deck raises both questions at once. Precision is what you ask looking into the net: of everything in here, how much is fish and how much is weed and old boots? Recall is what you ask looking back at the sea: of all the fish that were down there, how many are now on deck? A small net cast carefully can be pure and still leave the sea full of fish. A net that spans the harbour catches everything and comes up mostly rubbish.

Now notice what neither formula contains. The true negative box, the one that swallowed accuracy in the town of a thousand, appears in neither. Precision and recall are computed entirely from the positive class and the mistakes around it, which is exactly why they keep working when the negatives outnumber the positives a thousand to one.

That fishing net also contains a suggestion. If a bigger net raises recall and lowers precision, then something must be adjustable, and adjusting it must move both.

One dial sets them both, and it moves them opposite ways

You now have two measures that disagree, and nothing yet that says how to trade one against the other. The instrument that does it has been hidden all along, and the next five figures move it from one end of its range to the other. A model does not really answer yes or no. Underneath, it produces a **score**: a number saying how strongly this case looks positive to it. A score of 0.93 means it is fairly convinced. A score of 0.31 means it is not.

A yes-or-no answer appears only when somebody draws a line and says: at or above this, call it positive. That line is the **threshold**, and it is not part of the model. It is a decision made afterwards, by a person, and it can be moved at any time without retraining anything.

The eight passengers are set aside here only because a threshold needs more cases to show its shape; they return in the section after next. Figures 1.24 to 1.27 use twenty four patients, laid out left to right by the score the model gave them, with the eight who genuinely have the condition on the upper row. The model is decent but not clairvoyant: it ranks most of the eight high and puts a few healthy people up there too.

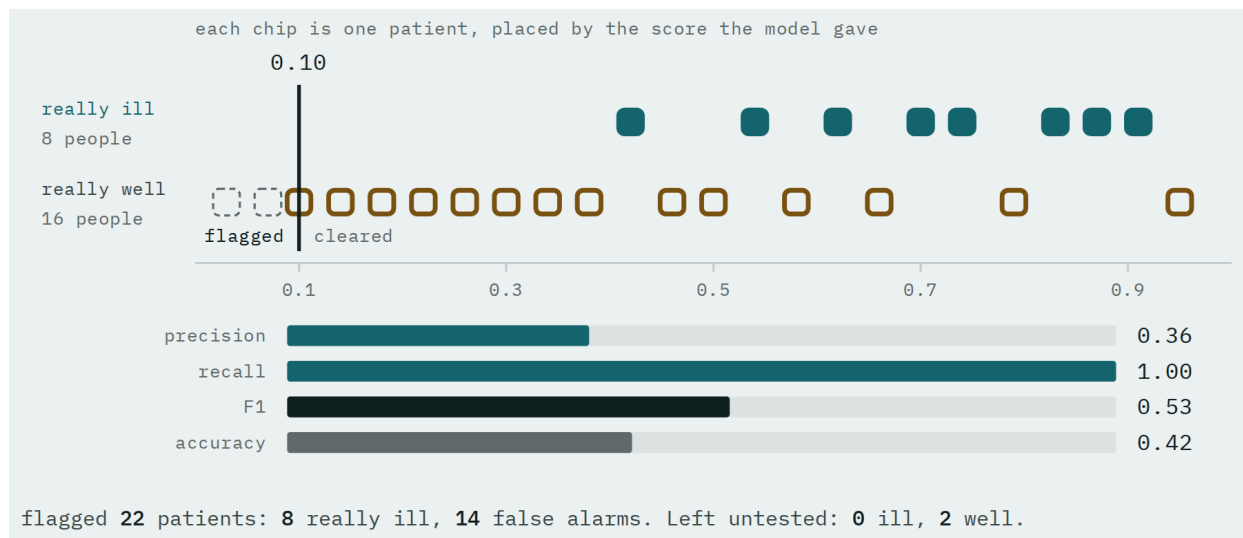


Figure 1.24 The threshold at 0.10. Twenty two patients flagged: all eight who are ill, and fourteen who are not. Precision 0.36, recall 1.00.

At the loosest setting the model flags almost everybody. It catches all eight patients, so recall reaches 1.00. It also drags in fourteen healthy people, so precision collapses to 0.36. Every ill person is caught and most of the alarms are worthless.

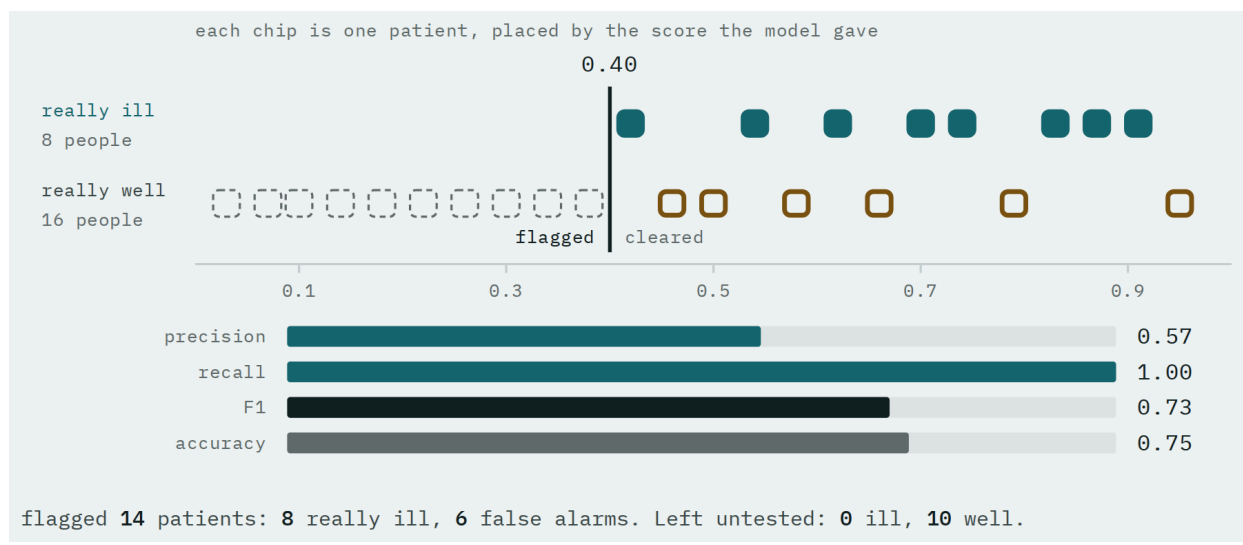


Figure 1.25 The threshold at 0.40. Fourteen flagged, still all eight ill, six false alarms. Precision 0.57, recall 1.00.

Raising the line to 0.40 discards six false alarms without losing a single ill patient. Precision climbs to 0.57 and recall has not moved. This is the part of the range where tightening costs nothing, and it does not last.

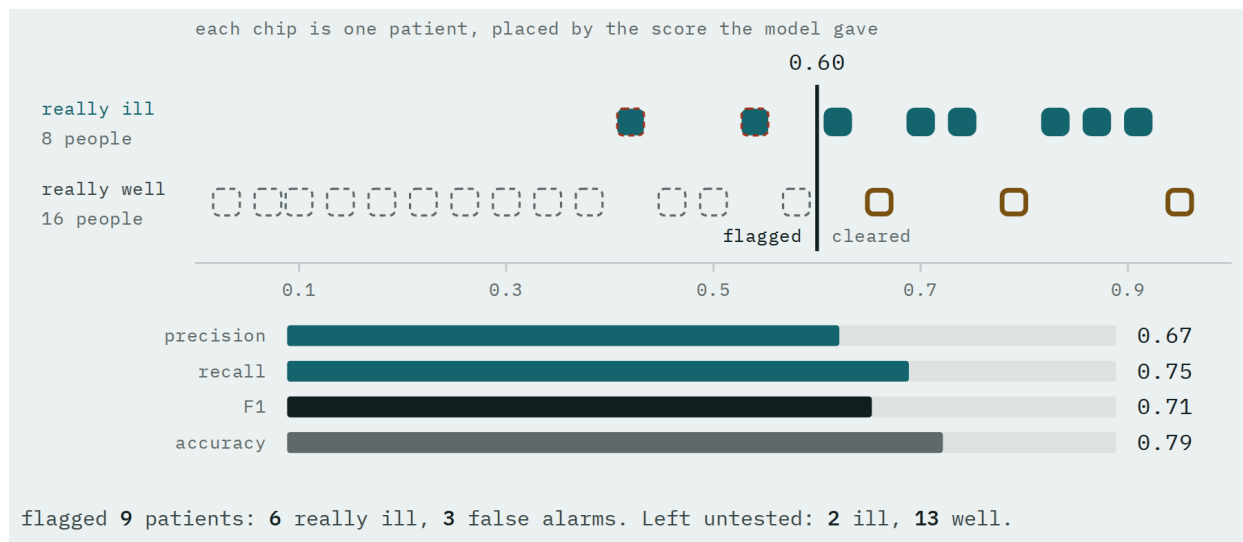


Figure 1.26 The threshold at 0.60. Nine flagged: six ill and three false alarms. Precision 0.67, recall 0.75.

Two ill patients have now been lost. Recall falls to 0.75 and precision continues to rise. From here every gain in precision is paid for in missed cases.

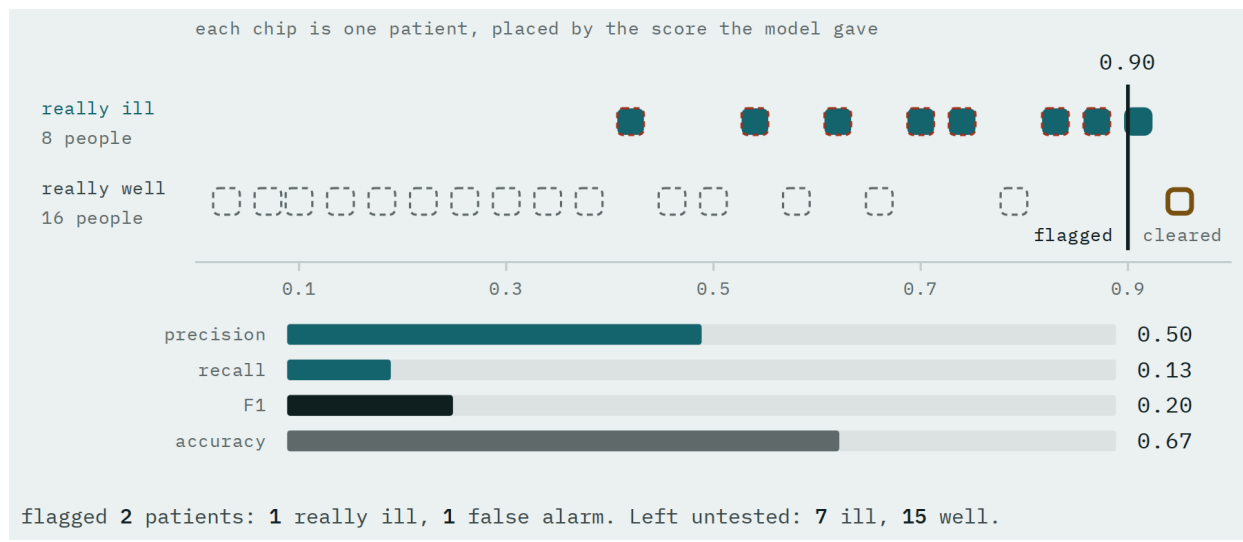


Figure 1.27 The threshold at 0.90. Two patients flagged, one of them ill. Recall falls to 0.13.

At the strictest setting only two people are flagged, and seven ill patients walk out untested. Recall has fallen to 0.13.

This is the trade-off, and it is not a flaw in any particular model. It is arithmetic. Flagging more cases can only add to the flagged pile, so it can only help recall and can only hurt precision. No threshold makes both perfect unless the model separates the two groups completely, and real models do not.

Airport security decides how sensitive to make the walk-through arch. Turn it up and belt buckles set it off: nothing dangerous gets past, and the queue fills with people emptying their pockets. Turn it down and the queue flows, almost every alarm is worth investigating, and something eventually walks through. The dial does not change what is in anyone's bag. It chooses which of the two failures you would rather have today.

Accuracy is worth watching across this range too. It sits highest between 0.60 and 0.70, barely moves across the middle, and drops sharply only at the loosest setting, because with sixteen healthy patients out of twenty four it is still mostly reporting on the healthy ones. Precision and recall are the two that actually move.

Two numbers moving in opposite directions is awkward for a leaderboard. So people built a third to fold them together, and the way they folded them was deliberate.

F1 refuses to let a strong half cover for a weak one

If one setting gives high recall and another high precision, you need a single number to rank them by. Try the obvious fix first. Precision 0.36 and recall 1.00: average them and you get 0.68. That model flags twenty two of the twenty four patients and is wrong about most of its alarms, and the plain average has just called it fairly good.

Worse, compare it with the line at 0.70, where precision is 0.71 and recall 0.63. That model is balanced and useful, and its plain average is 0.67, which is lower. The ordinary average prefers the one that flags nearly everybody.

The **F1 score** uses a different kind of average, the **harmonic mean**. Instead of adding the two numbers and halving, it multiplies them, doubles the product, and divides by their sum.

$$F1 = 2 \times (\text{precision} \times \text{recall}) / (\text{precision} + \text{recall})$$

line at 0.10 $F1 = 2 \times (0.364 \times 1.000) / 1.364 = \mathbf{0.53}$ *plain average 0.68*

line at 0.70 $F1 = 2 \times (0.714 \times 0.625) / 1.339 = \mathbf{0.67}$ *plain average 0.67*

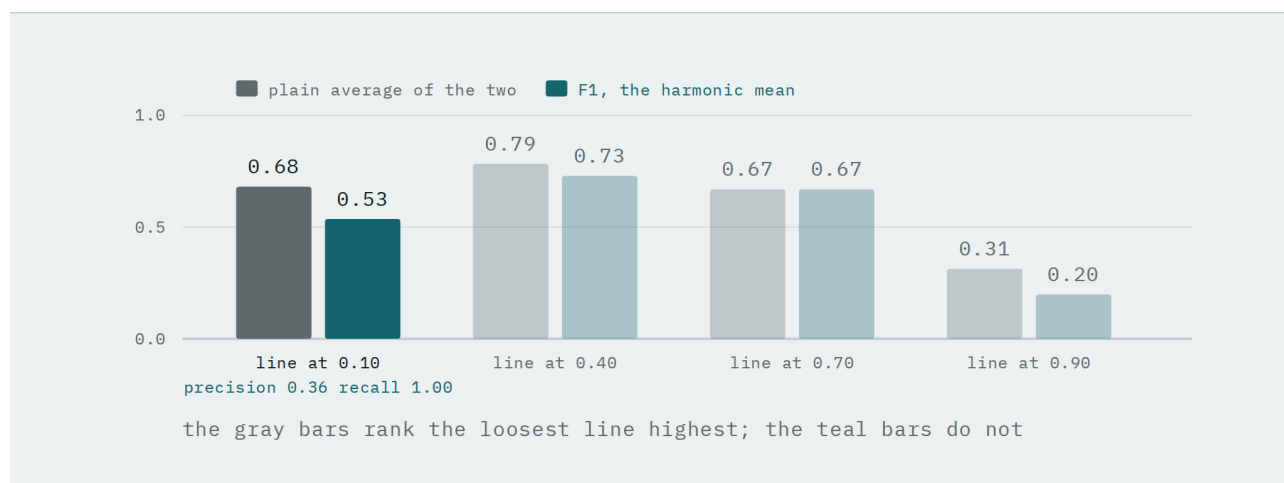


Figure 1.28 The two averages at four settings, with the loosest line selected. The plain average rates it 0.68; F1 rates it 0.53.

In Figure 1.28 the gray bar and the teal bar disagree most sharply, at the setting where the model is least discriminating. Figure 1.29 selects the balanced line instead.

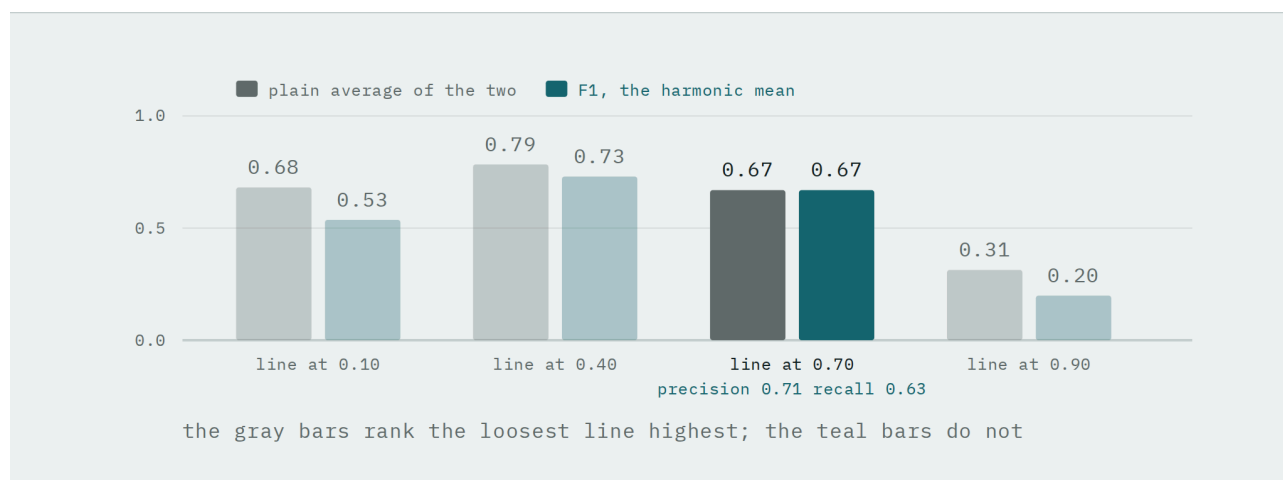


Figure 1.29 The same four settings with the line at 0.70 selected. Precision 0.71 and recall 0.63 give a plain average of 0.67 and an F1 of 0.67.

Compare the two figures and the ranking flips. By plain average the loose setting wins, 0.68 against 0.67. By F1 the balanced setting wins decisively, 0.67 against 0.53. Only one of those two verdicts matches what you would want from a screening programme.

The harmonic mean sits close to the smaller of the two numbers, and the further apart they are the harder it pulls down. Any value multiplied by something near zero is near zero, and precision and recall are multiplied together in the numerator. So one number near zero drags the whole score there however good the other one is.

A driving test with a theory paper and a road test makes the same choice. Score 100 on the theory and 20 on the road, and averaging says 60, a pass. Nobody runs a driving test that way, because the two halves are not interchangeable and being brilliant at one does not make you safe.

The extreme case makes it plainest. Push the line below anything the range offers, so that every single patient is flagged, and recall is 1.00 with precision 0.33, because a third of the patients are ill. The plain average is 0.67, a respectable-sounding number for a model that does no work at all. The F1 is 0.50, and if the illness were rarer that would fall further.

The 1 in F1 is a weight. It says precision and recall matter equally, which is a real assumption and often the wrong one. When they do not matter equally the family is called F-beta: F2 weights recall higher, F0.5 weights precision higher. Reporting plain F1 quietly claims that missing a case and raising a false alarm cost you about the same.

The same square read one row at a time gives four rates

TPR, FNR, TNR and FPR. Four more abbreviations, and not one of them is a new idea. Each is a single box divided by the row it sits in, as Figures 1.30 and 1.31 show one row at a time.

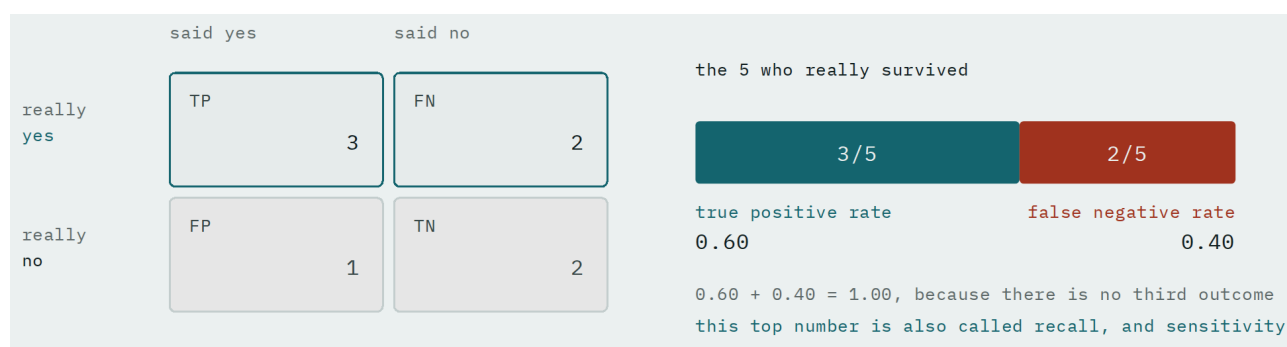


Figure 1.30 The top row, the five who really survived, split into a true positive rate of 0.60 and a false negative rate of 0.40.

Figure 1.30 takes the top row: everyone who was really positive, five survivors. Of those five, three were caught and two were missed. Three fifths is the **true positive rate**, 0.60. Two fifths is the **false negative rate**, 0.40. They add to 1, and they must, because every genuine survivor was either caught or missed and there is no third option.

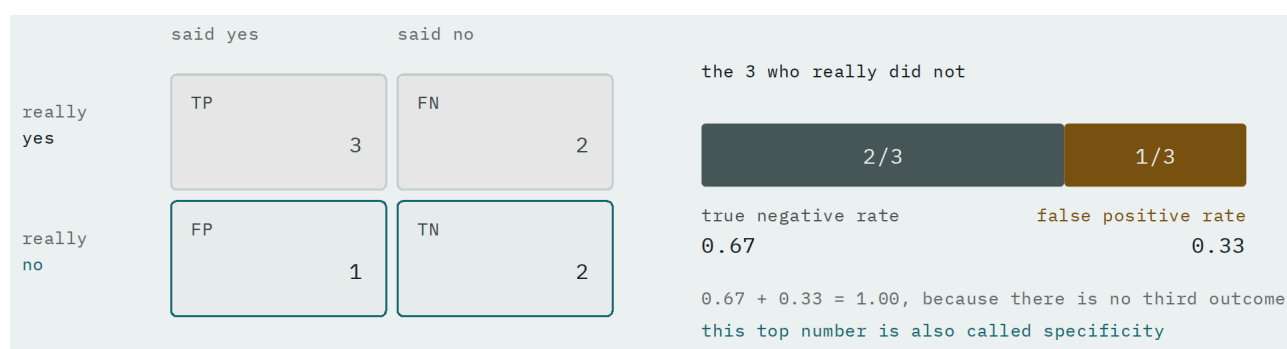


Figure 1.31 The bottom row, the three who really did not survive, split into a true negative rate of 0.67 and a false positive rate of 0.33.

Figure 1.31 takes the bottom row: everyone who was really negative, three deaths. Two were correctly left alone, giving a **true negative rate** of 0.67, and one was wrongly flagged, giving a **false positive rate** of 0.33. Those two also add to 1, for the same reason.

Now the part worth reading twice. The true positive rate is three fifths. Recall is three fifths. They are the same number under two names, because they are the same division: the true positive box over the row of real positives. Medicine has a third name for it, **sensitivity**, and the true negative rate has a fourth, **specificity**. One recipe printed in four cookbooks under four titles. A reader who has cooked it once will follow any of the four; a reader meeting all four at once assumes there are four dishes.

These are the rates you meet in clinical papers, and there is a good reason for it. Sensitivity and specificity are computed inside a row, so they do not depend on how common the disease is in the room. Precision is computed down a column, mixing the two rows together, so the same test reports a very different precision in a screening clinic than in a specialist ward. That is the effect Figure 1.20 was showing.

Choosing a measure is deciding which mistake you can afford

There is no best measure, and a chapter that ended by naming one would be misleading you. There is only a question you have to answer first, in words, about your own situation: which of the two mistakes hurts more, and roughly by how much?

Answer that and the measure follows almost mechanically. Refuse to answer it and no formula will answer it for you, because the arithmetic has no way of knowing what a missed tumour costs relative to an unnecessary biopsy. Figures 1.32 to 1.34 set out three situations where the answer differs.



Figure 1.32 Screening for cancer. A missed case costs far more than a false alarm, so recall leads.

A false negative sends a sick person home reassured. A false positive sends a well person for a scan they did not need. Those costs are nowhere near equal, so recall leads and a great deal of precision is spent buying it. Everyone who screens positive goes on to a second, sharper test, and that design only makes sense once you have accepted a low precision on the first one.



Figure 1.33 A spam filter. A false alarm costs more than a missed message, so precision leads.

Here it inverts. A false negative is a junk mail you delete in a second. A false positive is a job offer in a folder nobody opens. Precision leads, and some spam is allowed through on purpose.

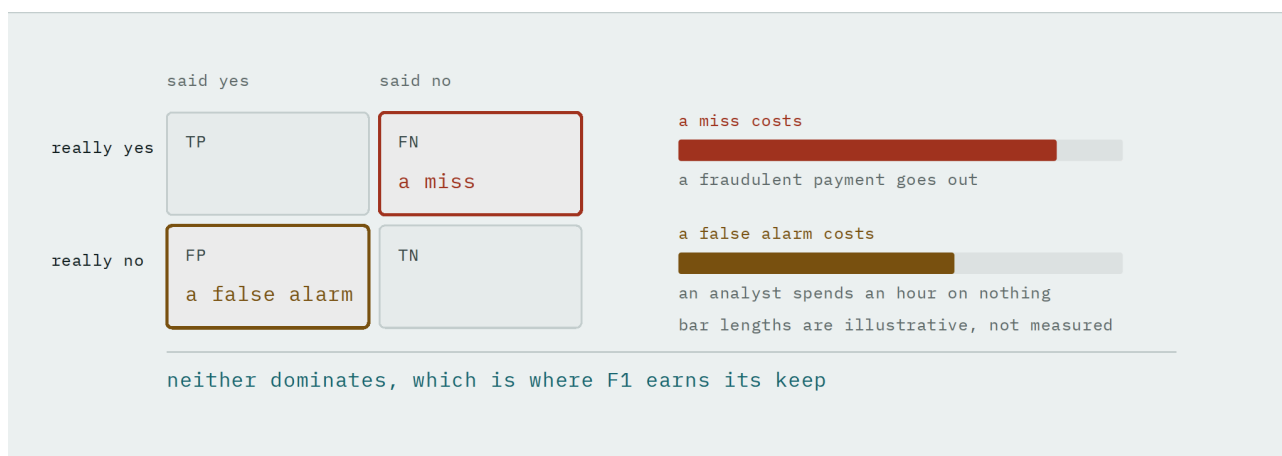


Figure 1.34 A fraud review queue. Neither error dominates, which is where F1 earns its keep.

Analysts can only work through so many cases a day, so a flood of false positives wastes the team, and every missed fraud is money gone. Neither error dominates and the queue length itself sets the threshold.

A fire alarm and a bank vault alarm cannot be tuned the same way. The fire alarm should scream at burnt toast, because the cost of missing a fire is a life and the cost of a false alarm is a corridor full of irritated people. A vault alarm that sends armed response every time a lorry rumbles past becomes an alarm nobody answers. Same device, opposite settings, and the settings come from what the mistake costs rather than from anything inside the device.

You should also know where these four numbers stop being trustworthy. They describe one model, on one dataset, at one threshold, for one definition of positive. They say nothing about whether the model works on a different hospital's patients, whether the labels were correct to begin with, or whether the cases were a fair sample. A confusion matrix built on a test set that leaked into training reports beautiful numbers about nothing at all. And a recall of 0.60 computed on eight people is not a property of the model. It is one noisy sample of one, and a different eight could easily read 0.4 or 0.8.

Everything in this chapter came out of one square with four numbers in it. Accuracy pools all four and gets fooled when one box is enormous. Precision reads a column and recall reads a row. F1 refuses to let one of them carry the other, and the four rates are those same rows written as fractions. The only judgement that was never in the arithmetic is which box you most need to keep small.

The square counts what happened. Deciding which corner of it you can live with is your job, not the model's.